

Matthew Hase-Liu

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MIT mathematician in algebraic geometry and number theory, with experience in research-level LLM evaluation and fine-tuning open-weight models. Interested in reproducible post-training, evaluation, and inference-time methods for improving language-model reasoning and reliability, with a particular interest in open models and mathematical applications.

Experience

Mathematics Research Collaborator Aug. 2026–Present

Google DeepMind via Mercor · Project details confidential

- Generate synthetic mathematical data for language-model training.
- Evaluate model-generated solutions and proofs in research-level mathematics.
- Investigate methods to improve the clarity and precision of model-generated mathematical exposition.

Research Scientist and NSF Postdoctoral Fellow Aug. 2026–Present

Massachusetts Institute of Technology · Mentor: Bjorn Poonen

- Developing novel variants of the circle method to better understand moduli spaces and rational points.

Machine Learning Intern Winter 2017, Summer 2018

Otter.ai

- Modified and trained an LSTM punctuation-prediction model to add capitalization support.
- Tested clustering refinements to improve speaker diarization in a production transcription pipeline.

Selected ML Project

Math Journal Suggester Code · Experiment · Demo

- Built a tool that suggests journals for a math paper from its title and abstract: Qwen3-Embedding-8B retrieves 20 of 95 journals, and a Kev-4B reranker orders them. Self-hosted the demo on an NVIDIA GB10.
- Fine-tuned Kev’s LoRA adapter and decision head on 8,000 papers. On 1,000 held-out papers, off-the-shelf Kev did no better than retrieval (39.6% vs. 40.6% top-3); fine-tuning raised top-3 to **72.4%** and top-1 from 21.1% to **51.3%**.

Education

Ph.D. in Mathematics May 2026

Columbia University · Advisor: Will Sawin

Research: moduli spaces of curves on hypersurfaces via the circle method.

S.M. in Computer Science May 2021

Harvard University

A.B. in Mathematics May 2021

Harvard College · Summa cum laude with highest departmental honors

Selected Mathematical Research

1. The asymptotic in Waring’s problem over function fields beyond twice the degree (2026). *Submitted*. Established the **optimal threshold** $n > 2d$ for the uniform asymptotic with a power-saving error, under explicit field hypotheses.
2. Birch’s theorem over function fields with quadratically many variables (2026). *Submitted*. Developed a new geometric method that reduces the required number of variables from **exponential to quadratic** in the degree.
3. A geometric approach to functional equations for general multiple Dirichlet series over function fields. *Algebra & Number Theory*, to appear.
4. A higher genus circle method and an application to geometric Manin’s conjecture. *Algebra & Number Theory*, to appear.

Full publication list · 18 invited talks and posters, including at Harvard, MIT, and Stanford.

Awards

NSF Mathematical Sciences Postdoctoral Research Fellowship (2026–30) · NSF Graduate Research Fellowship (2021–26)
Putnam, Rank 107.5/4638 (2018) · USA Math Olympiad, Rank 37/280 (2017) · Phi Beta Kappa (2021)

Technical Skills

Programming: Python, PyTorch, Hugging Face (Transformers, PEFT, Accelerate), SageMath, Linux

Languages: English (native), Japanese (fluent)